

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

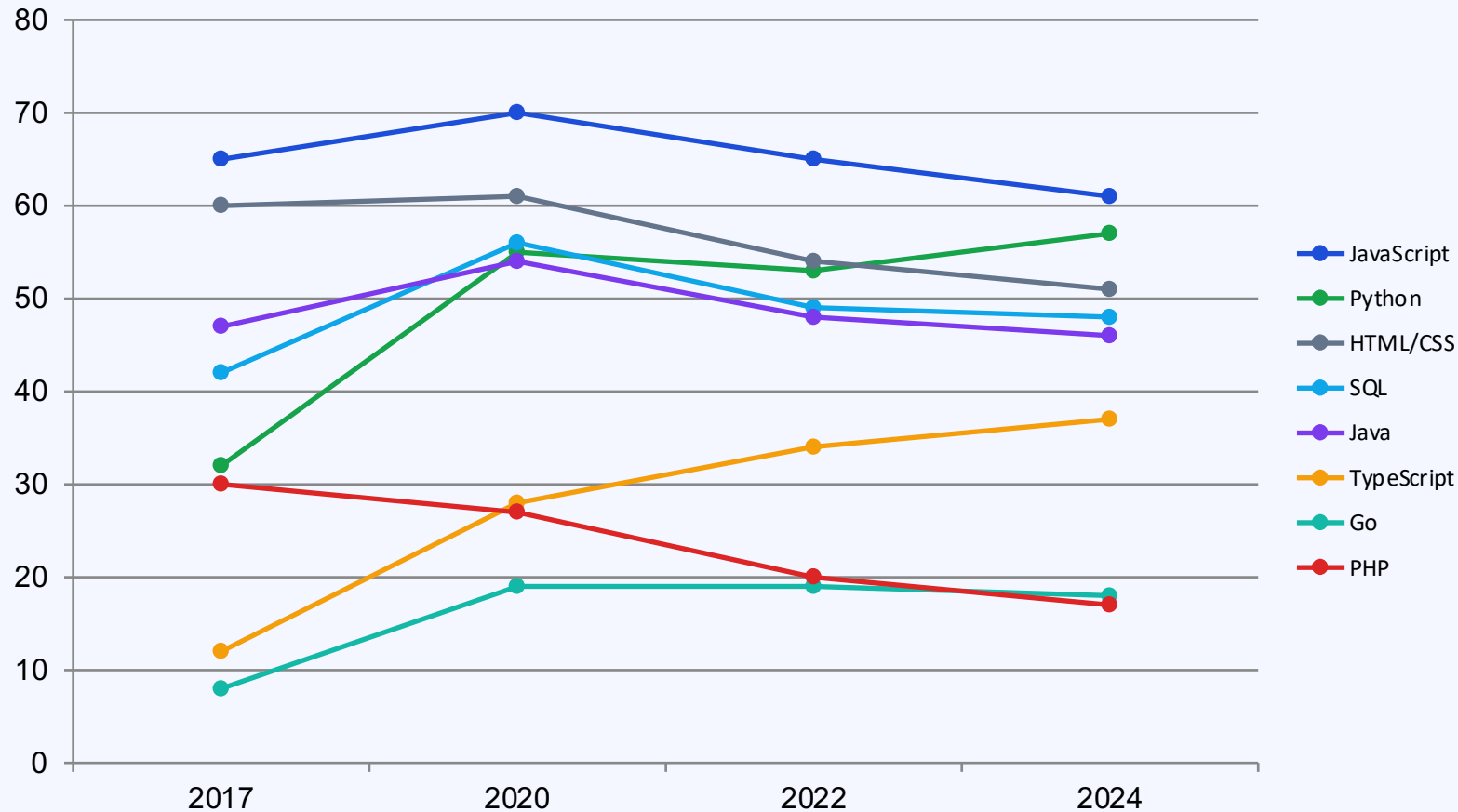
JetBrains annual survey of 23,262 developers worldwide

- Most comprehensive annual snapshot: languages, tools, AI attitudes, and career trends
- Focus: usage shifts (2017–2024), developer preferences, and domain-language fit
- Audience: engineering leaders, CTOs, and developer experience teams

Narrative: How usage is changing → what developers prefer → what to do about it

Python Closes the Gap: 8 Years of Language Usage Shifts

Usage share (% of developers), 2017–2024

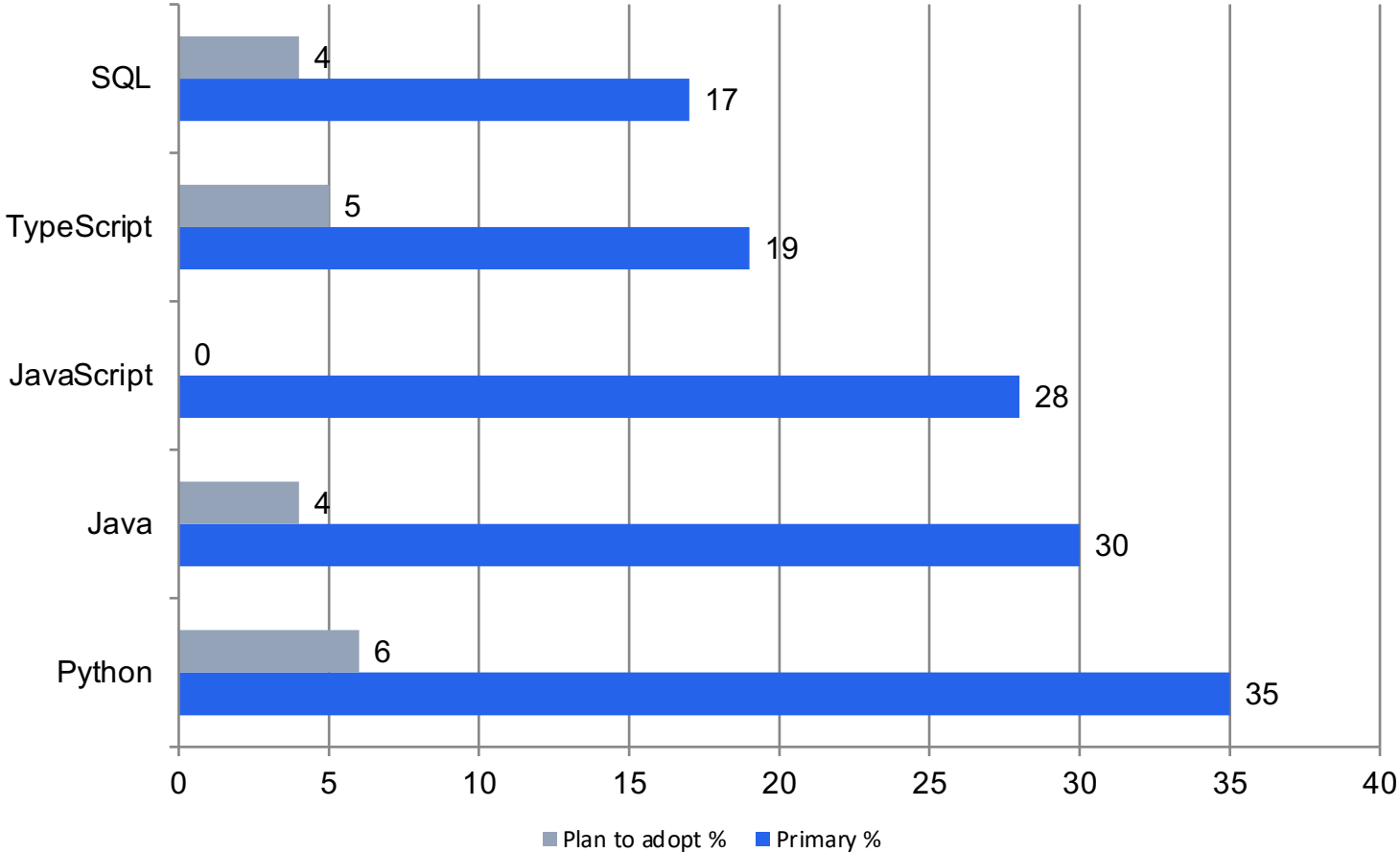


Key trend signals

- JavaScript: 70% peak (2020) → 61% (2024)
- Python: 32% → 57%, near parity with JS
- TypeScript: 12% → 37% (3x growth)
- PHP: 30% → 17% sustained decline
- Top-5 stack is being reshuffled

Python Is Now the #1 Primary Language at 35%

Primary language share vs. planned adoption as primary



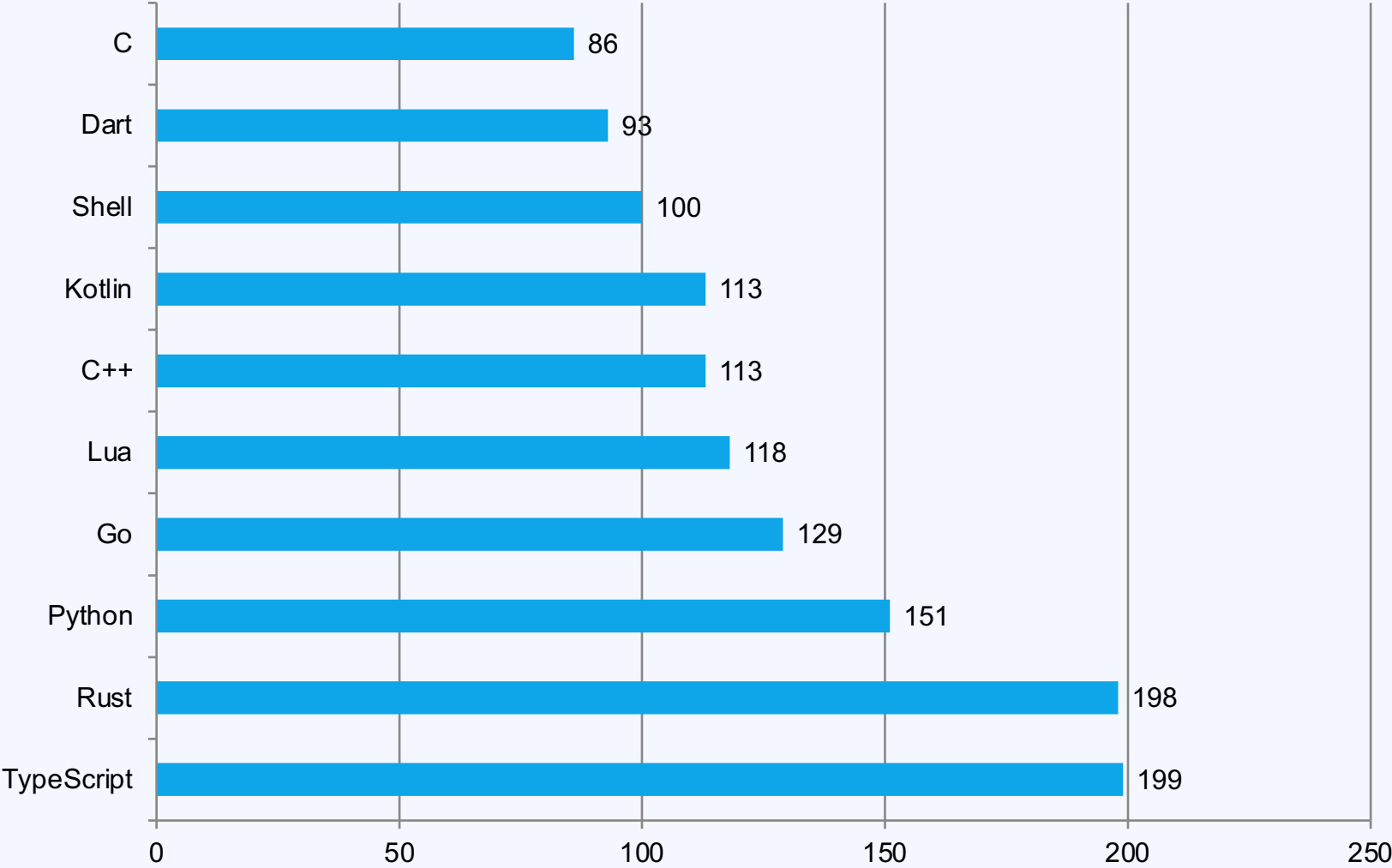
Ecosystem lock-in

Adoption intent stays below 6% across top languages, indicating high switching costs and strong stack familiarity.

Implication

Prioritize upskilling and internal mobility over external language-switch hiring assumptions.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



What stands out

- TypeScript (199) and Rust (198) are effectively tied at the top
- Top ranks skew toward type-safe, modern-tooling ecosystems
- Python remains highly liked while keeping broad usage breadth

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

Interpretation

Language strategy should be domain-led, not one-size-fits-all platform standardization.

Notable pattern

Python dominates AI/data; TypeScript/JavaScript own UI; Java and C++ stay strong in backend and systems.

What This Means for Your Technology Strategy in 2025

1 Invest in Python for AI/ML teams

Python leads AI/ML (31%) and data science (28%), with strongest primary-language position (35%).

2 Standardize on TypeScript for frontend/full-stack

TypeScript dominates UI (62%) and leads satisfaction (199), supporting velocity and developer experience.

3 Develop Rust/Go capability for infrastructure

Rust and Go combine adoption momentum with strong satisfaction for reliability-critical systems.

4 Plan for shrinking legacy hiring pools

PHP declined from 30% to 17% (2017→2024); Ruby also declined from 10% to 4%.

5 Use tooling quality as a retention lever

Higher satisfaction clusters around type safety and modern tooling; prioritize this in platform strategy.

Bottom line: Optimize language strategy by domain fit + developer satisfaction, not historical defaults.